

Business Process Automation

with Power Automate and Copilot Studio Agents

WSQ Course Code: TGS-2022017524 · 2-Day Hands-On Training · Modules 1–3

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W
Version 3.0 (3 Jul 2026) · www.tertiarycourses.com.sg · <https://lms-tms.tertiaryinfotech.com/>

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

CERTIFICATION

Microsoft Power Platform, AI and data analytics — 20+ years of training experience.

DELIVERS

WSQ courses on AI, business automation, data science and software engineering.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Let's Know Each Other

- Your name and organisation / role.
- Your experience with Power Automate, Copilot Studio or other automation tools (if any).
- One repetitive business process you would love to automate after this course.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

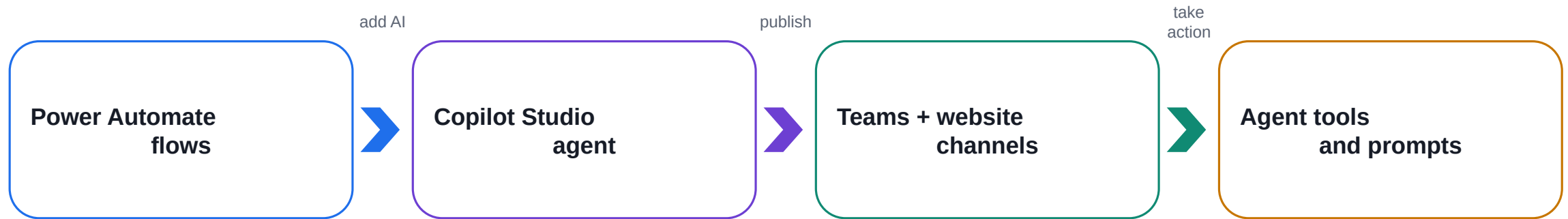
5

Be punctual; return from breaks on time.

6

75% attendance is required.

What You'll Learn



Day 1 — automate

Build reliable trigger → action flows without an AI agent.

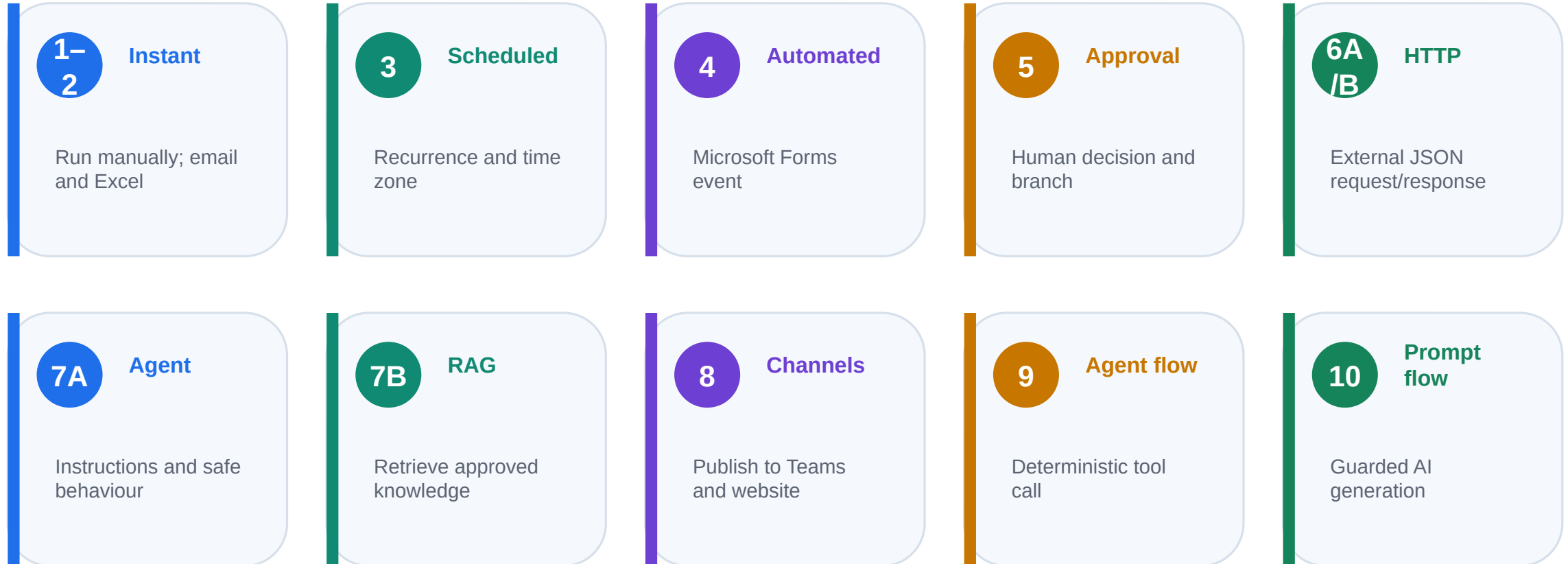
Day 2 — converse

Create a grounded agent that understands and confirms user intent.

Integrate — act

Let the agent call deterministic or prompt-based Power Automate tools.

Your 2-Day Journey



Each lab reuses the previous capability; complexity increases one controlled step at a time.

How This Course Works

Concept, then demo, then hands-on — for every topic

Built entirely in Microsoft 365 and the Power Platform

No coding required — low-code, drag-and-drop tools

Save your flows and agents to reuse in your own role

- Daily recap and open Q&A to lock in what you learned

SCHEDULE

Lesson Plan — 2 Days, 9:00am–6:00pm

Day 1

- Day 1 — Foundations & Power Automate
 - Modules 1–2: workflow concepts, Power Automate
 - Labs 0–5: email, Excel logging, approval, scheduled, form flows

Day 2 & assessment

- Day 2 — Business Agents with Copilot Studio
 - Module 3: agents, prompts, agent + flow
 - Labs 6–11: first agent, knowledge (RAG), tools, agent flows
- Assessment (Day 2, 4:00–6:00pm)
 - Written Assessment 1 hr + Practical Performance 1 hr
- Daily timing
 - 9:00am–6:00pm · 1-hour lunch · tea breaks within

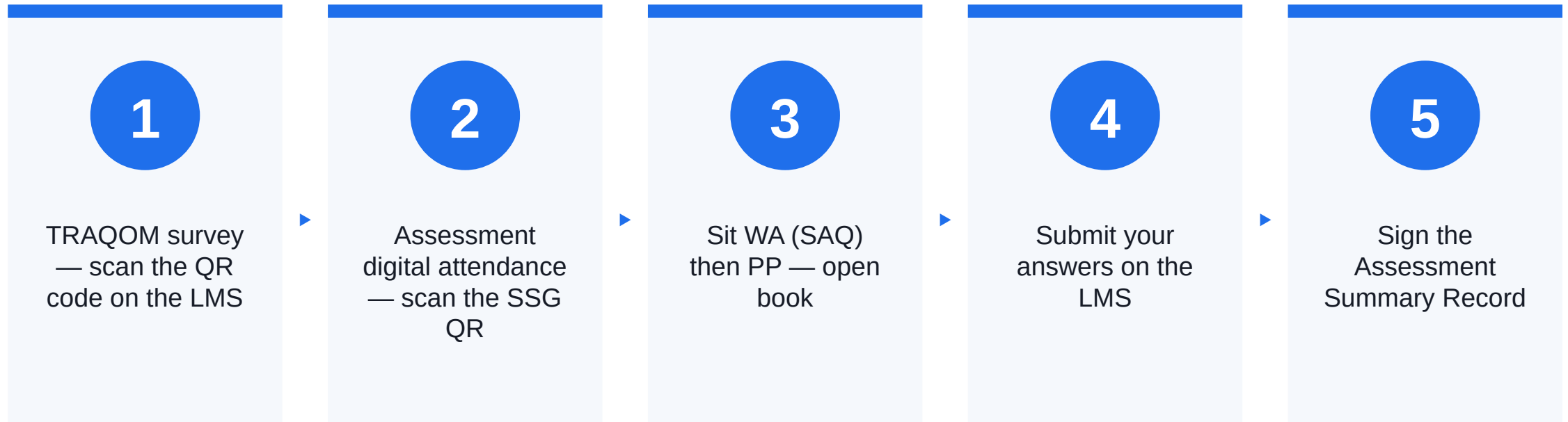
Briefing for Assessment

- Place phones and other materials under the table or on the floor.
- No photos or recording of assessment scripts.
- No discussion during the assessment.
- Use a black/blue pen for hard-copy assessments.
- No liquid paper / correction tape.
- Scripts are collected when time is up.

Assessment & Funding

- Written Assessment (SAQ) — 1 hour, open book. Tests the knowledge from Modules 1–3.
- Practical Performance (PP) — 1 hour, open book. You build a working flow + agent, as in the labs.
- Both are taken on Day 2, 4:00–6:00pm, and submitted on the LMS.
- WSQ funding requires 75% attendance AND passing the assessment (Competent).
- Courseware and the assessment are on the LMS: <https://lms-tms.tertiaryinfotech.com/>

Assessment Flow



Access the Hands-On Labs

All 12 labs (Labs 0–11) + the Learner Guide live in the course GitHub repository:

github.com/tertiarycourses/TGS-2022017524---Business-Process-Automation-with-Power-Automate-and-Copilot-Studio-Agents

Option A — Clone with Git

- Install Git (git-scm.com) if needed.
- `git clone <repo URL>.git`
- Open the labs/ folder in your editor or browser.

Option B — Download the ZIP

- Open the repo URL in your browser.
- Code → Download ZIP, then unzip.
- Open labs/ and follow each lab's index.md.

Day 1 — Foundations & Power Automate

Modules 1 & 2 · Understand automation, then build your first flows

Module 1 · Introduction to Workflow Automation

Why automate, and the building blocks behind every workflow

What Is Business Workflow Automation?

A workflow is a repeatable series of steps that gets work done

Automation lets software run those steps for you

It replaces manual, repetitive tasks — copying data, sending emails, chasing approvals

Runs reliably and consistently, with far fewer errors

- People focus on judgement; software handles the routine

Why Automate?

Save time — reclaim hours of manual work every week

Reduce errors — consistent, rule-based execution every time

Respond faster — instant notifications and approvals

Gain visibility — every step is logged and traceable

- Scale easily — handle more volume without more headcount

Manual Process vs. Automated Workflow

MANUAL PROCESS

Emails get missed in busy inboxes

Data re-keyed by hand between apps

Approvals stall and get forgotten

No audit trail of what happened

- Slow, inconsistent, hard to scale

AUTOMATED WORKFLOW

Triggers fire the moment an event happens

Data flows automatically between systems

Approvals routed, tracked and reminded

Every action logged automatically

- Fast, consistent and scalable

Standalone Agents vs. Integrated Workflows

STANDALONE AGENT

Answers questions and chats

Works inside the conversation

Output stays in the chat

- Great for information and guidance

INTEGRATED WORKFLOW

Takes action across systems

Connects apps, data and people

Produces results: records, emails, approvals

- Great for getting work done

Introduction to Workflow Logic

Every workflow follows the same simple pattern

Trigger → Action(s) → Output

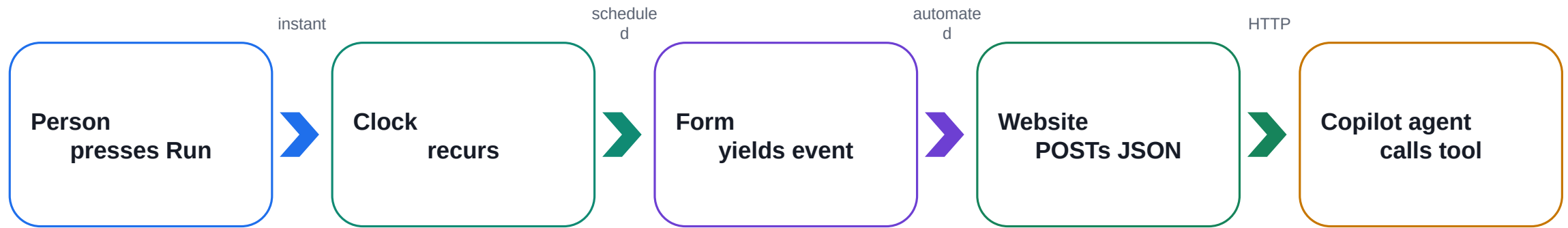
Trigger: the event that starts the workflow

Actions: the steps the workflow performs

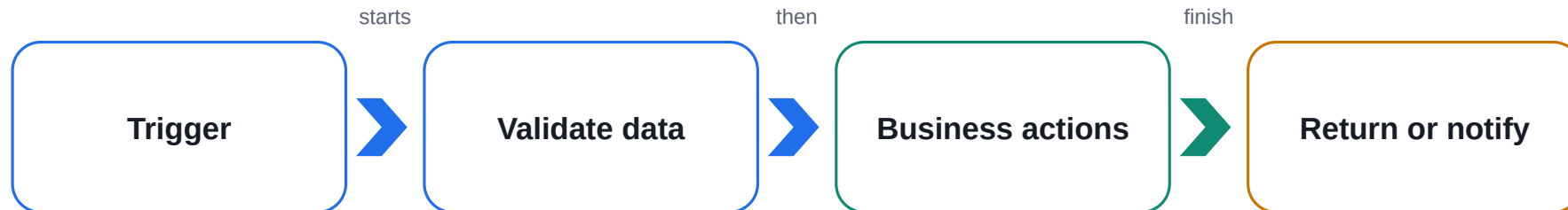
Output: the result the workflow produces

- Master this pattern and you can build any automation

Triggers — What Starts a Workflow



One trigger starts a flow. The actions may stay the same even when the trigger changes.



Actions — What the Workflow Does

Actions are the steps performed after the trigger fires

Examples: send an email, add a row to Excel, post a notification, request an approval

They run in sequence — each step can reuse data from earlier steps

Combine many actions to model a complete business process

- Add conditions to decide which actions run

Outputs — The Result

The output is the outcome the workflow produces

A logged record, a sent email, an approval decision, a notification

Outputs can become the input to the next workflow

- Judge success by the output: did the right thing happen, reliably?

Workflow Steps — Putting It Together

Steps run top-to-bottom after the trigger

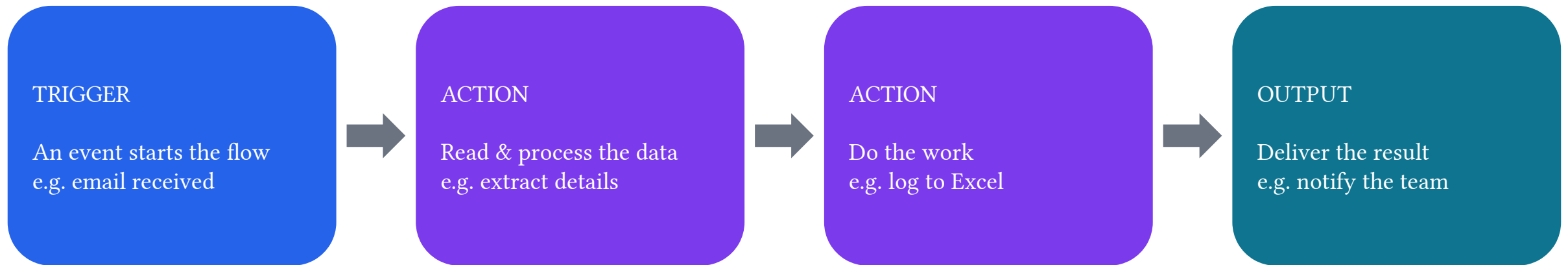
Add conditions (if / then) to branch the logic

Pass data between steps using dynamic content

Example: New email → read the details → log to Excel → notify the team

- Build complexity gradually — start simple, then extend

Anatomy of a Workflow



Master this pattern — Trigger → Action(s) → Output — and you can build any automation.

Module 1 Recap

Automation runs repeatable work for you — saving time and cutting errors

Standalone agents inform; integrated workflows take action

Every workflow follows: Trigger → Actions → Output

Triggers start it, actions do the work, outputs deliver the result

- Next: build real flows in Microsoft Power Automate

Module 2 · Introduction to Power Automate

Triggers, actions, and your first five working flows

Overview of Power Automate

Microsoft's low-code tool for automating tasks and processes

Connects 1,000+ apps and services — Outlook, Excel, SharePoint, Teams, Forms and more

Build flows visually: pick a trigger, add actions, done

Runs in the cloud — no servers, no code to maintain

- Part of Microsoft 365 and the Power Platform

Types of Flow and Process Patterns

1 Instant

A person presses Run

Labs 1–2

2 Scheduled

A clock reaches a timetable

Lab 3

3 Automated

An event occurs

Lab 4

4 Agent flow

A Copilot agent calls a tool

Labs 9–10

Human-in-the-loop pattern

Lab 5 pauses a running flow until a person approves or rejects.

HTTP request pattern

Labs 6A–6B let an external website POST JSON and receive JSON back.

The Maker Experience

Build flows at make.powerautomate.com

Designer canvas: trigger at the top, actions added below

Search a connector, choose an action, fill in the fields

Insert dynamic content to pass data between steps

- Test, then turn the flow on — it runs automatically from then on

Common Triggers

A trigger is the event that starts your flow

Email received — react to incoming mail in Outlook

File upload — a new file lands in SharePoint or OneDrive

Form submission — someone responds to a Microsoft Form

- Each trigger hands its data to the rest of the flow

Trigger: Email Received

Starts a flow when a new email arrives in Outlook

Filter by sender, subject, folder or importance

Grabs the email's data: from, subject, body, attachments

Use it for: enquiries, requests, alerts and notifications

- Lab example: a sales enquiry email kicks off logging and a reply

Trigger: File Upload

Starts a flow when a file is added to SharePoint or OneDrive

Watches a specific folder or document library

Captures the file name, content and metadata

Use it for: invoices, contracts, reports and submissions

- Lab example: an uploaded invoice starts an approval workflow

Trigger: Form Submission

Starts a flow when someone submits a Microsoft Form

Each answer becomes data you can use downstream

No inbox or folder to watch — the form is the front door

Use it for: requests, registrations, feedback and orders

- Lab example: a purchase request form routes to a manager for approval

Creating Workflow Actions

After the trigger, add the steps that do the work

Search for a connector, then pick the action you need

Fill in the fields — type text or insert dynamic content

Stack actions in order; reorder by dragging

- Three actions you'll use constantly: send email, add Excel row, request approval

Action: Send Emails

Sends an automated email from Outlook

Set the recipient, subject and body — any can be dynamic

Personalise with data from the trigger (sender name, request details)

Use for confirmations, notifications and auto-replies

- Keeps everyone informed without anyone lifting a finger

Action: Create Excel Entries

Adds a new row to a table in an Excel workbook

Map each column to data from the flow

Builds a running log automatically — enquiries, requests, transactions

Stored in OneDrive or SharePoint, ready to filter and report on

- Turns scattered events into a clean, structured record

Human in the Loop — Approval Pattern



If yes — Approve

Send the approved message and continue the business process.

If no — Reject

Send the rejected message and stop or return for changes.

The automation is still deterministic; the person supplies the decision.

Connectors — Linking It All Together

A connector is the bridge to an app or service

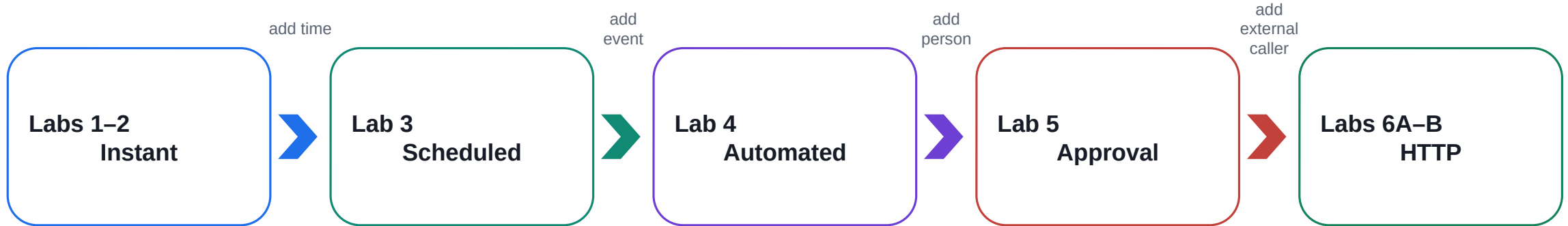
Outlook, Excel, SharePoint, Teams, Forms, Approvals and 1,000+ more

Each connector offers its own triggers and actions

Sign in once; the flow uses your secure connection

- Mix connectors freely to span your whole toolset

Hands-On Labs — Power Automate



DAY 1 BUILDS THE AUTOMATION ENGINE

Trigger

What starts the process?

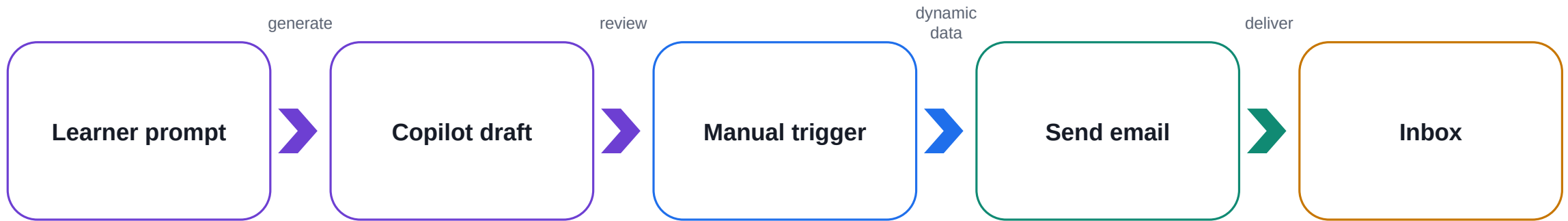
Actions

What work should happen?

Outputs

What result must be stored or returned?

Lab 1 — Instant Email Flow



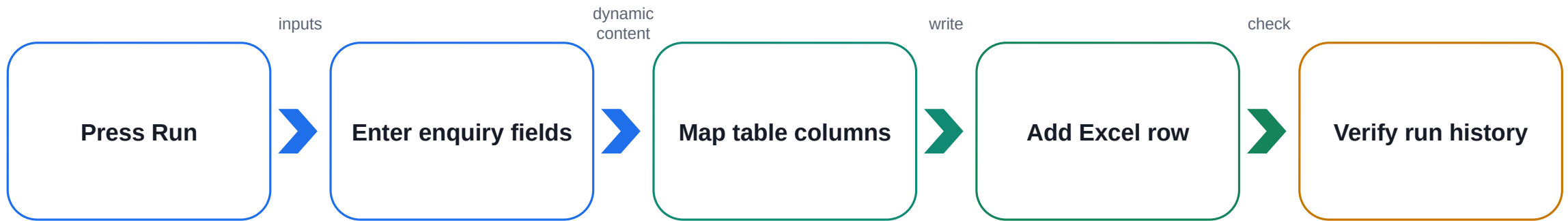
Concept

An instant flow runs only when a person deliberately starts it.

Workplace use

A service officer acknowledges a verified phone or counter enquiry within 15 minutes.

Lab 2 — Instant Excel Logging Flow



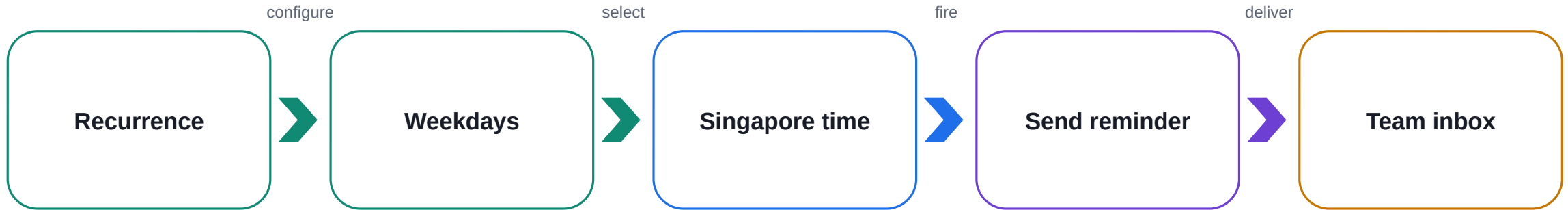
Concept

Actions consume outputs from the trigger as structured dynamic content.

Workplace use

The shared enquiry register gives supervisors one visible queue and a basic audit trail.

Lab 3 — Scheduled Flow



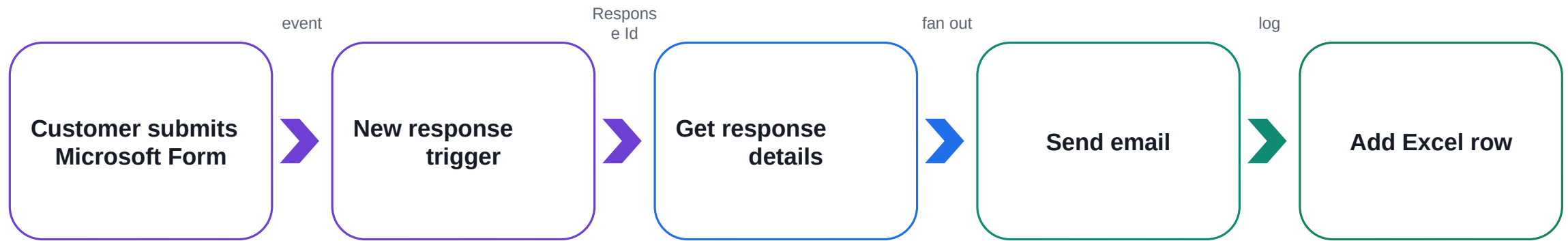
Workplace use

At 9 AM on weekdays, the team lead reminds officers to clear records still marked New.

Critical setting

Always set the business time zone; recurrence defaults can otherwise use UTC.

Lab 4 — Automated Form Flow



Workplace use

A customer form creates the service email and audit row without staff re-keying.

Data pattern

Get response details turns the response ID into fields that email and Excel can use.

Labs 5–6 — Human and External Triggers

5

Lab 5 — Laptop approval

SGD 1,850 request
↓
Manager decides
↓
Approve / Reject
↓
Notify requester

6A

Lab 6A — Website enquiry

Business-banking form
↓
POST JSON
↓
Notify operations
↓
Confirm on page

6B

Lab 6B — Service widget

Customer question
↓
Approved routes
↓
Safe fallback
↓
Return bot reply

Day 1 Recap — Choose the Right Trigger

Instant

Person presses Run

Labs 1–2

Learn actions and data mapping

Scheduled

Recurrence

Lab 3

Run by the clock

Automated

Forms event

Lab 4

React to a business event

Approval

Flow pauses for a person

Lab 5

Add human judgement

HTTP

External JSON request

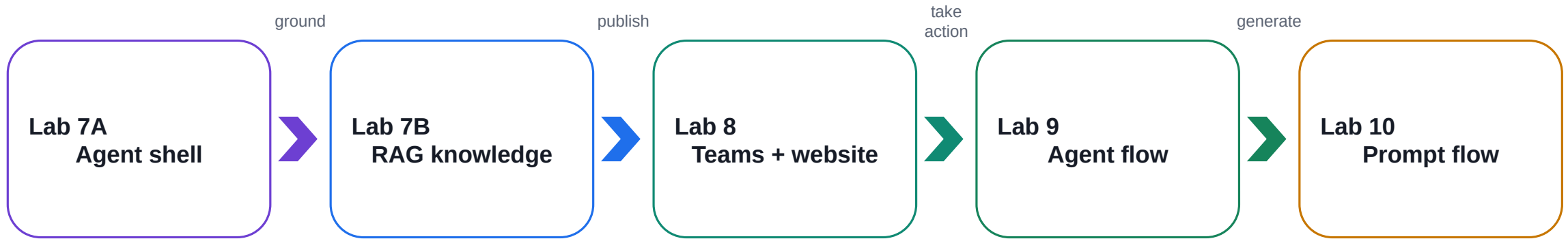
Labs 6A–B

Connect a website or system

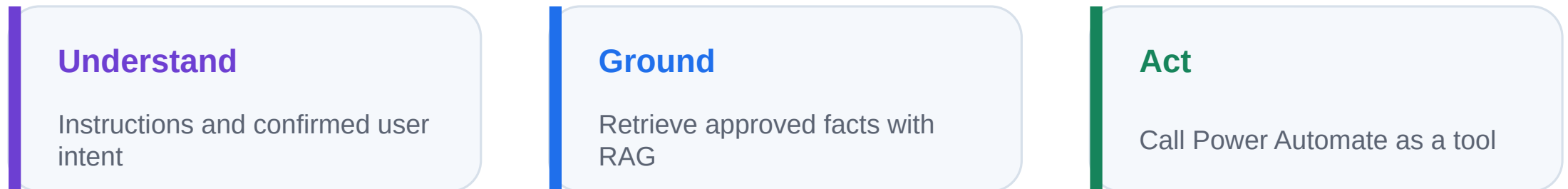
Day 2 — Building Business Agents with Copilot Studio

Module 3 · Create AI agents and connect them to your flows

Day 2 Roadmap



DAY 2 ADDS THE CONVERSATIONAL ORCHESTRATOR



Module 3 · Building Business Agents with Copilot Studio

From a helpful chatbot to an agent that triggers real work

What Is Copilot Studio?

Microsoft's low-code tool for building custom AI agents

Create conversational agents that understand natural language

Ground them in your own knowledge — documents, sites, data

Agents can call Power Automate flows to take action

- Publish to Teams, websites and other channels

The Copilot Studio Maker

Build agents at copilotstudio.microsoft.com

Create a blank agent, then edit its details and instructions

Add knowledge sources, topics and tools

Test live in the built-in chat panel as you build

- Publish when you're happy — update any time

How an Agent Works

Instructions tell the agent its role and how to behave

Knowledge gives it the information to answer from

Topics handle specific conversations — the agent chooses one from its description

Tools let it do things — including running your flows

- It interprets the user, responds, and can hand off to automation

What Makes a Good Business Agent

A clear, narrow job — it does one thing well

A defined audience and tone of voice

Reliable, structured output the next step can use

Grounded in trustworthy knowledge — no guessing

- Knows when to act: hand off to a flow at the right moment

Creating Practical Business Agents

Start from a real task: 'handle sales enquiries', 'log procurement requests'

Write clear instructions: role, scope, what to collect, what to avoid

Add knowledge: product info, policies, price lists, FAQs

Decide the output: what should the agent hand back?

- Test with realistic questions before connecting any automation

Prompt Design Fundamentals

The prompt is the instruction that shapes the agent's response

Be specific: state the role, the task and the format you want

Give examples of good answers — the agent learns the pattern

Set boundaries: what to do when information is missing

- Iterate: test, refine the wording, test again

Prompt Design for Structured Outputs

A workflow needs predictable fields, not free-flowing prose

Ask the agent to return the same fields every time

Name each field clearly: Customer, Product, Quantity, Priority

Specify the format — a labelled list, a table, or JSON

- Structured output = data your Power Automate flow can read directly

Structured Output Formats

LABELLED FIELDS

Customer: Acme Ltd

Product: Widget Pro

Quantity: 50

Priority: High

- Easy to read, easy to map

JSON

```
{  
  "customer": "Acme Ltd",  
  "product": "Widget Pro",  
  "quantity": 50,  
  "priority": "High"  
}
```

- }

Example: A Structured Prompt

THE PROMPT

“You are a sales enquiry assistant.

From the customer's message, extract:

Customer name, Product, Quantity,
and Priority (High / Medium / Low).

Return the result as labelled fields.

- If a detail is missing, write ‘Not stated’.”

THE OUTPUT

Customer: Acme Ltd

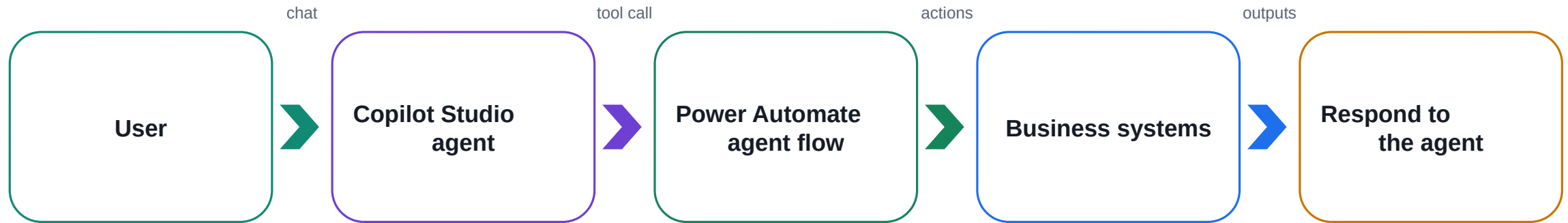
Product: Widget Pro

Quantity: 50

Priority: High

- → Ready to log to Excel and notify sales

Connecting Agents to Power Automate



Agent = brain and mouth

Understands, asks, confirms and chooses a tool.

Flow = hands

Logs, emails, approves, calls APIs and applies rules.

Return value = answer

Reference, decision or draft comes back into chat.

Passing Outputs into Workflows

The agent's structured output becomes the flow's input

Each field maps to a flow parameter: Customer, Product, Quantity

The flow uses those values in its actions — log, email, approve

Clean structured output makes mapping effortless

- Garbage in, garbage out — good prompts pay off here

Input and Output Mapping

Define the flow's inputs to match the agent's fields

Map each agent field to the matching flow parameter

The flow can return a value back to the agent (e.g. a reference number)

The agent confirms to the user: 'Logged — your reference is 1042'

- Data flows both ways: agent → flow → agent

Best Practices for Agent + Flow

Keep each flow focused on one clear job

Validate inputs — handle missing or odd data gracefully

Use clear field names shared between agent and flow

Test the flow on its own before wiring it to the agent

- Tell the user what happened — confirm success or explain failure

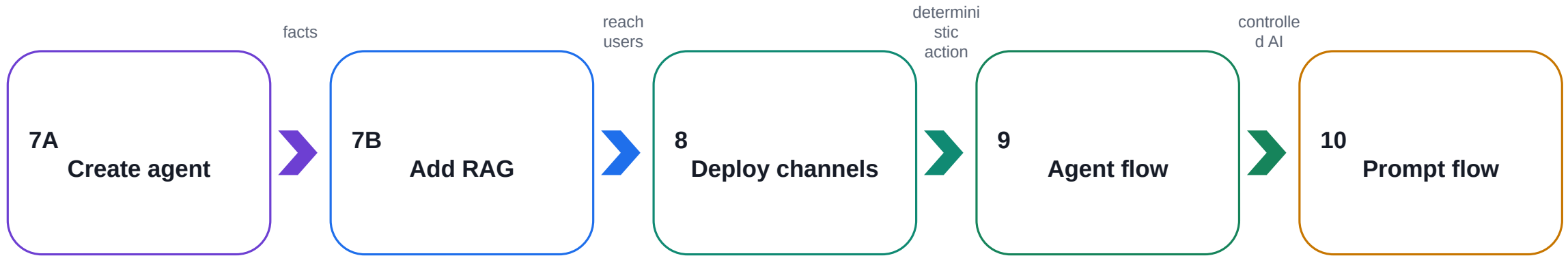
n8n AI Agent Pattern in Microsoft

n8n AI-agent pattern

Microsoft equivalent



Hands-On Labs — Copilot Studio



Build or configure

Create the agent and inspect every generated instruction.

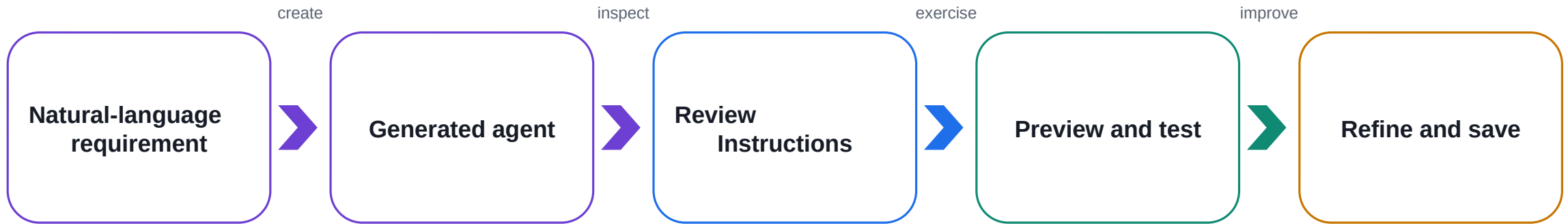
Test evidence

Use positive, negative and unsupported questions.

Publish safely

Validate in Preview before Teams or website channels.

Lab 7A — Create the IT Support Agent



Workplace role

First-line IT support for password, MFA, VPN and service-desk escalation.

What it does not have yet

No approved internal facts and no Power Automate tool.

Lab 7B — Ground the Agent with RAG



Evidence found

Give the approved VPN, password or security procedure and cite the FAQ.

No authority

Refuse access approvals and route the employee to the Service Desk.

Lab 8 — Deploy to Teams and Website

A Part A — Copilot agent in Teams

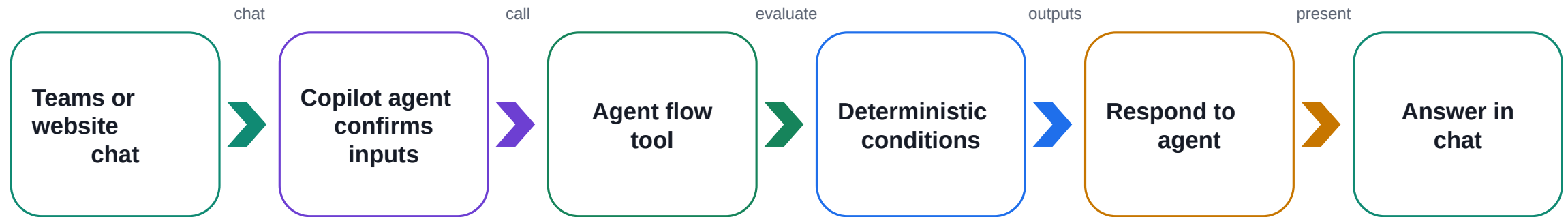
User chats in Teams
↓
Copilot Studio agent
↓
Grounded conversation
↓
Answer in Teams

B Part B — Website calls deterministic flow

Website enquiry form
↓
HTTP POST
↓
Deterministic Power Automate
↓
JSON result on page

Comparison: Part B is automation only—the agent is not in that submission path yet.

Lab 9 — Agent Calls a Deterministic Flow



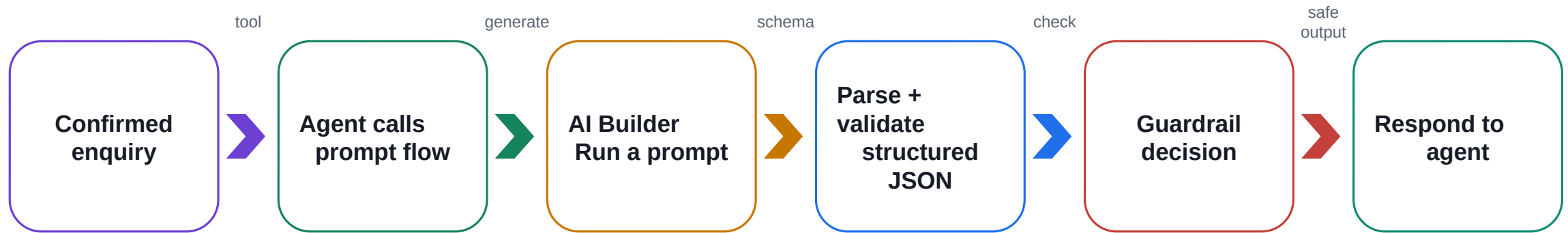
Same as n8n AI Agent + workflow tool

The agent decides when to call the tool; the flow executes predictable business rules.

Required endpoints

When an agent calls the flow → actions → Respond to the agent.

Lab 10 — Agent Calls a Prompt Flow



Use AI where language varies

Draft, classify or summarise—but keep approvals, identity checks and limits deterministic.

Validate before returning

Parse the schema, check safety rules and fall back when the output is missing or unsafe.

Compare Labs 8, 9 and 10

8

Lab 8 — Ordinary flow

TRIGGER
Website HTTP request

PROCESSING
No agent in path

RETURN
JSON to webpage

9

Lab 9 — Agent flow

TRIGGER
Agent tool call

PROCESSING
Deterministic rules

RETURN
Structured result to chat

10

Lab 10 — Prompt flow

TRIGGER
Agent tool call

PROCESSING
AI prompt + guardrails

RETURN
Controlled draft to chat

Testing & Debugging Agents

Test in the chat panel with realistic, messy inputs

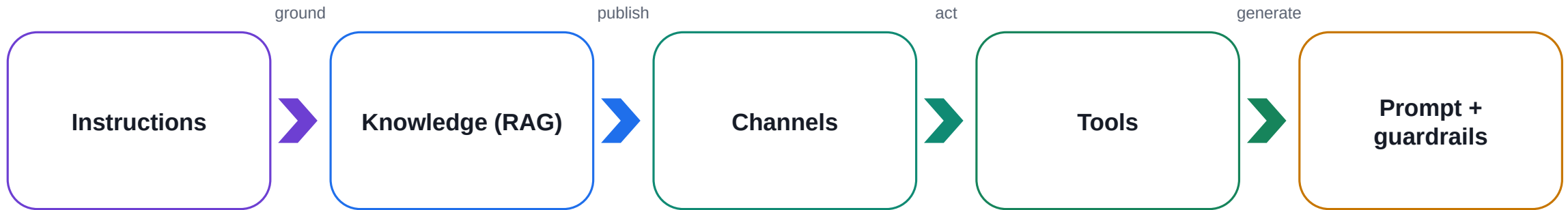
Check the output format is consistent every time

Trace flow runs to see exactly what data passed through

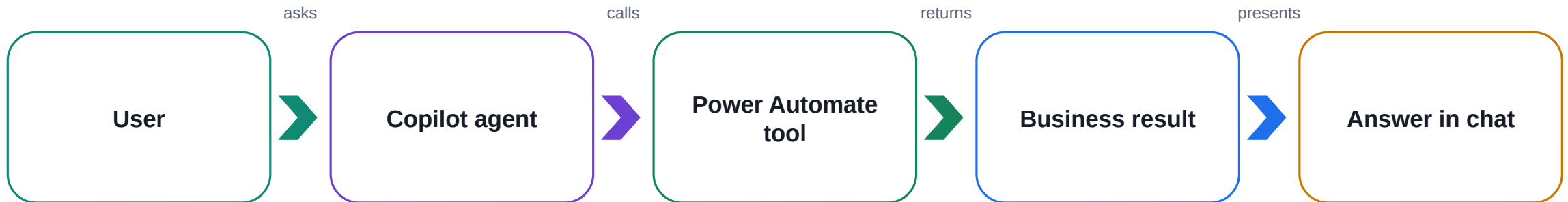
Refine the prompt when answers drift or fields go missing

- Handle the edge cases: missing info, unexpected questions

Day 2 Recap — The Agentic Architecture



THE FINAL ARCHITECTURE



What You Can Automate Next

Onboarding: new-hire checklists and welcome emails

Reporting: scheduled data pulls and summaries

Helpdesk: triage and route incoming requests

Document handling: filing, approvals and reminders

- Anywhere people copy data or chase approvals by hand

Common Pitfalls to Avoid

Automating a broken process — fix the process first

Trying to build everything in one giant flow

Skipping tests, then surprises in production

No error handling — failures go unnoticed

- Vague agent prompts that return inconsistent data

Common Errors & Quick Fixes

“Unauthorized” sending email — reconnect Outlook with a mailbox-enabled account

Approval ‘valid users’ error — pick the approver from the dropdown (a real user in your tenant)

Date shows literal utcNow() — enter it via the fx / Expression editor so it becomes a token

Excel cell shows ##### — the column is just too narrow; auto-fit it

- Unwanted ‘For each’, or agent can’t see its flow — use single-value fields; same environment in both tools

Governance & Security Basics

Use connections and permissions you're authorised to use

Keep sensitive data out of emails and logs where possible

Follow your organisation's data and approval policies

Document what each flow does and who owns it

- Review and retire flows that are no longer needed

Rolling This Out at Work

Pick one painful, repetitive task to automate first

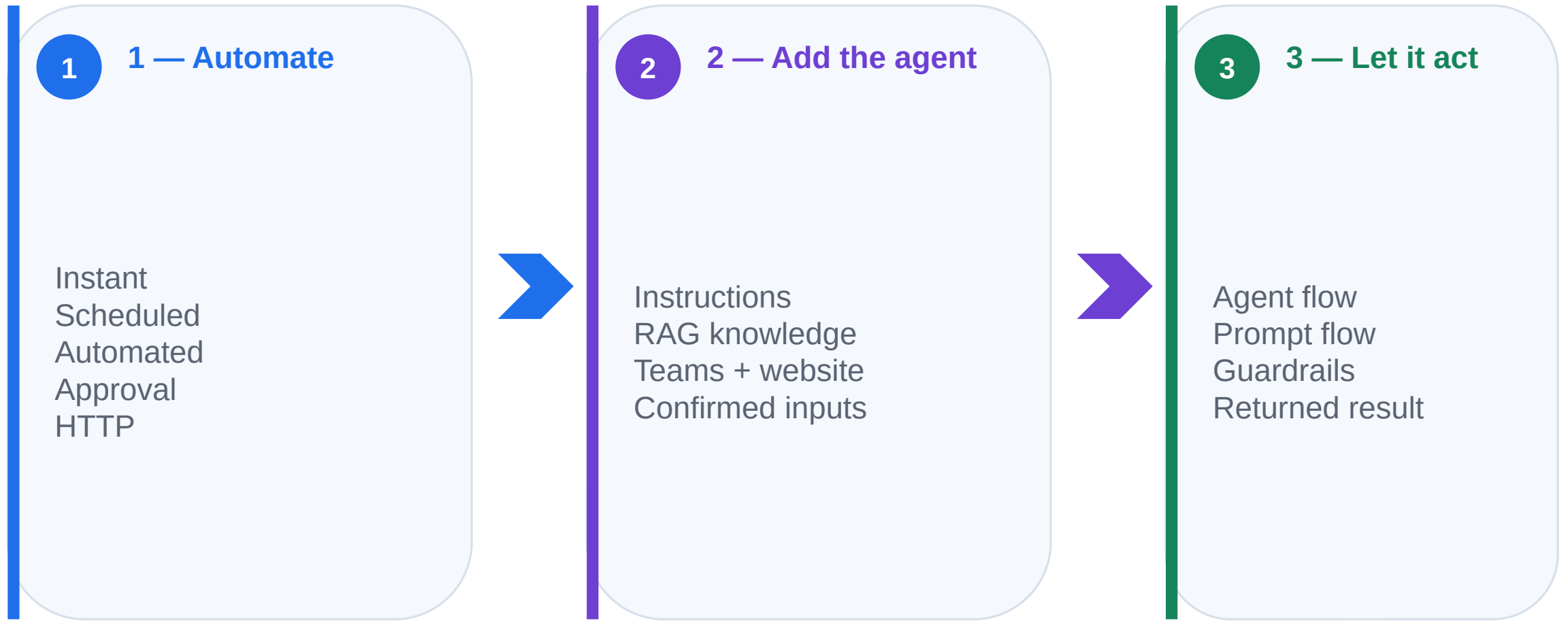
Prove the value, then share the win with your team

Build a small library of reusable flows and agents

Involve the people who do the work in the design

- Iterate — improve flows as needs change

Course Recap — The Big Picture



Key Takeaways

Every workflow is Trigger → Actions → Output

Power Automate does the work; Copilot Studio adds the intelligence

Structured agent outputs make flows reliable

Orchestration turns steps into complete processes

- Start small, test, and improve — then scale

Resources & Where to Go Next

make.powerautomate.com — build and manage your flows

copilotstudio.microsoft.com — create and publish agents

Microsoft Learn — free guided learning paths

Templates gallery — start from pre-built flows

- Your community — share flows and patterns with colleagues

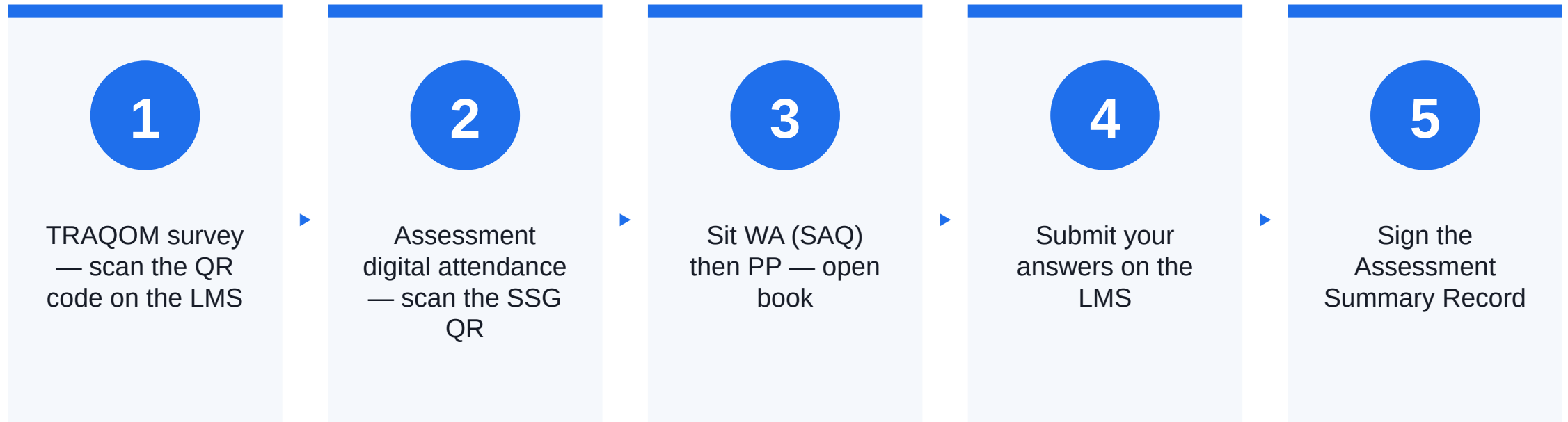
Courseware & Assessment on the LMS

- Download the slides and the Learner Guide from the LMS for the open-book assessment.
- Portal: <https://lms-tms.tertiaryinfotech.com/>
- Submit your Written Assessment and Practical Performance answers on the LMS.
- Complete the TRAQOM survey via the QR code on the LMS.

Assessment

- Written Assessment (SAQ) — 1 hour. Practical Performance (PP) — 1 hour.
- Open book: slides, Learner Guide and approved materials only.
- Remember to take the Assessment digital attendance (TRAQOM).
- Submit your completed answers on the LMS at <https://lms-tms.tertiaryinfotech.com/>.

Assessment Flow



Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Thank You

Now go automate something — questions welcome